

Lecture 2: Vector Equation, $Ax=b$, and Solution Sets

Handout Notes

Overview

This second lecture comprises three sessions.

- Session 1: Revisit of previous lecture, homework issues, vector equations
- Session 2: $Ax=b$
- Session 3: Solution Sets

1 Vector Equations

Matrix A matrix is a **rectangular** array of numbers. If A has m rows and n

The ultimate goal of linear algebra is to **solve a system of linear equations**

- **Equation:** a mathematical statement that asserts the equality of two expressions
- **System:** a collection of one or more equations involving the same set of variables
- **Solution:** find all values of the unknowns that satisfy every equation in the system

Linear Equation: A linear equation in the variables $x_1, x_2, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b,$$

where b and the coefficients $a_1, a_2, \dots, a_n$ are real numbers.

Character of linear equations: ~~each term is either a constant or the product of a constant and a single variable (first power).~~ Maximum power of unknown is 1.

- $a_1x_1 + a_2x_2$ is linear
- $a_1x_1 + x_1x_2$ is not linear

Linear combinations preserve two properties: **additivity** and **homogeneity**.

- Additivity preserves addition, i.e., $f(x+y) = f(x) + f(y)$
- Homogeneity preserves scalar multiplication, i.e., $f(cx) = cf(x)$

1.1 Representation of Linear System

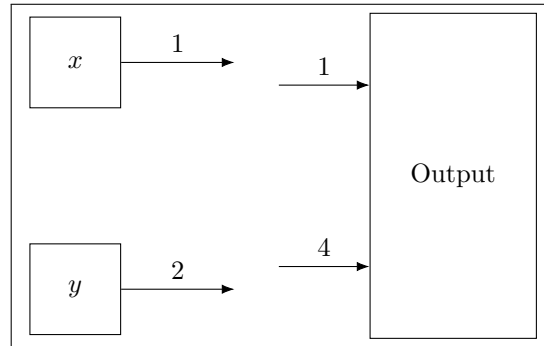


Figure 1: Basic Battery Manufacturing: Module X (Ore Machine) produces 1 unit of ore. Module Y (Gel Machine) produces 2 units of energy gel. Final input takes 1 ore and 4 gel to produce Battery Lv. I.

As shown in Figure 1, to find the solution of how many X and Y, we can solve either the equation of:

$$\begin{cases} 1x + 0y = 1, \\ 0x + 2y = 4, \end{cases}$$

whose solution is $(x, y) = (1, 2)$.

1.2 Three Pictures of Linear System

Consider the linear system

$$\begin{cases} 2x - y = 0, \\ -x + 2y = 3. \end{cases}$$

This is a system of two equations in two unknowns.

Row Picture In row picture, we depict the function image and find if there is an intersection point.

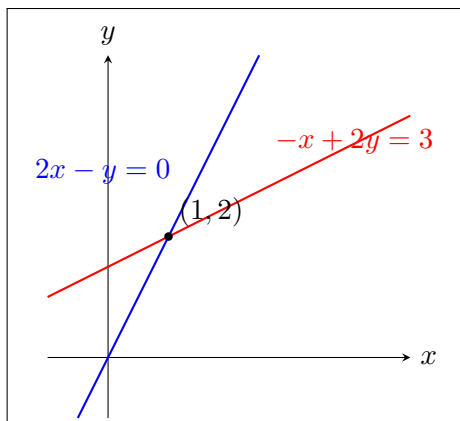


Figure 2: Row picture of the system $2x - y = 0$ and $-x + 2y = 3$.

Column Picture In column picture, we depict the vectors and combine them. After that, move around until we see the combined vector points to the $(0,3)$. The solution is the origin of the final vector.

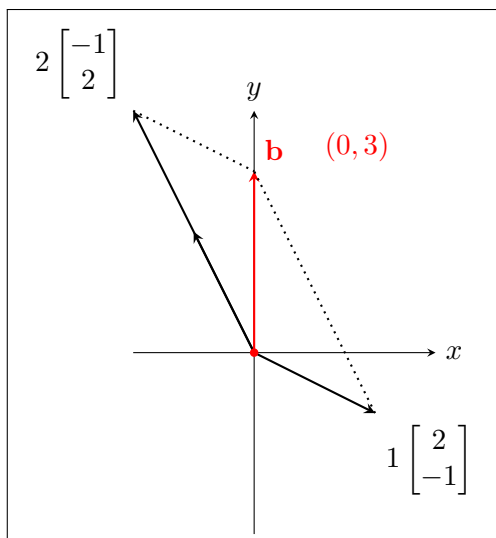


Figure 3: Column picture of the system showing the vector combination $1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$.

Matrix In matrix form, it is

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Or, we can write it as:

$$\left[\begin{array}{cc|c} 2 & -1 & 0 \\ -1 & 2 & 3 \end{array} \right].$$

That is the **augmented matrix**.

1.3 Properties of Linear Equation Systems

Dimension of System Linear systems may have n variables (columns) and m equations (rows). We name them $m \times n$ (linear) system.

Ideally, we like the ones where $m = n$, because it is *likely* to have a unique solution.

Think: why it is likely to have a unique solution?

Consistency of System A linear system is **consistent** if it has at least one solution. Otherwise, it is **inconsistent**.

Equivalent Systems Two linear systems are **equivalent** if they have the same solution set.

Think: what characterizes a linear system? What is the difference between consistency and equivalence?

1.4 Solution Sets

How many solutions can a linear system have?

- 1, when the lines intersect at a single point
- 0, for example, when the lines are parallel
- ∞ , when lines overlap

Think: is it possible for linear system to have 2 solutions?

2 Elimination

A system is characterized by its solution set. To solve a system, we perform elimination.

- **Input:** a system of linear equations.

- **Output:** an "Echelon form" system
- **Actions:** elementary row operations

2.1 Elementary Row Operations

There are three elementary operations that produces equivalent systems:

- Interchange: swap rows. Equations are not ordered
- Scaling: multiply a row by a (non-zero) constant. Linear systems preserve multiplication.
- Replacement: add a multiple of one row to another row. Linear systems preserve addition.

Think: how do we ensure the elimination to be feasible?

2.2 Coefficient Matrix

Recall that we discussed the three pictures of linear systems. Coefficient matrix is a compact representation of the system. Only the coefficients of variables are stored. The coefficient of the m^{th} equation's n^{th} variable is stored in the m^{th} row and n^{th} column of the matrix.

For the 3×3 linear system

$$\begin{cases} x + 2y - z = 1, \\ 2x - y + 3z = 7, \\ -x + 4y + 2z = 5. \end{cases}$$

Its coefficient matrix is

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \\ -1 & 4 & 2 \end{bmatrix}.$$

2.3 Augmented Matrix

We do need the numbers to the right of equal sign. When we add them to the right, we create a **augmented matrix**.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & -1 & 3 & 7 \\ -1 & 4 & 2 & 5 \end{array} \right].$$

2.4 Echelon Form

When performing elimination, we want to only keep x_1 on the 1st row (rest rows do not have x_1), only keep x_2 on the 2nd row (rest rows do not have x_2), and so on. For the system above, after elimination, we get:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -2.5 & 2.5 & 2.5 \\ 0 & 0 & 7 & 12 \end{array} \right],$$

we have zeros below the leading entries in each column. Consequently, this is called **Echelon Form**.

2.5 Pivot Elements

- Nonzero row or column: a row or column that contains at least one nonzero entry.
- Pivot column: leftmost nonzero column
- Pivot: nonzero entry at the top of pivot column
- Pivot row: row that contains a pivot

Basically speaking, because there can be many zeros in a row or column, we only care about, after the matrix has been sorted out, the first entry and corresponding entries afterwards.

2.6 Elementary Row Reduction

- Begin with leftmost nonzero column. This is a pivot column. The pivot position is at the top.
- Select the pivot (nonzero entry at the beginning). Let's start from the first unknown corresponding to x_1 , for example.
- Perform elementary row operation for all other entries in the column. As such, we eliminate all x_1 **except** for the selected row.
- Continue with column #2, #3, and so on.

Think: is echelon form unique?

2.7 How to Tell An Echelon Form

Well, as you may realize, not all systems are "beautiful". There are "talllllll" systems (Much more equations than unknowns), and "wiiiiiiiide" systems (Much more unknowns than equations).

Whatever it may be, remember that the basic idea of echelon form is to allow us tell what is the solution immediately.

Ideally, you can see a perfect "stair": if we forget zeros in the matrix, the first row starts from column 1, second row starts from column 2, ...

If the system is "talllllll" ($M > N$), then the last rows are all zeros, because the maximum non-zero rows is the number of unknowns.

If the system is wiiiiide ($N > M$), then there are columns not having leading entries.

For example:

$$\left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 4 \\ 0 & 0 & -1 & 2 & 3 \\ 0 & 0 & 0 & 3 & 1 \end{array} \right]$$

Here, the second variable x_2 is all 0. This is a **free variable**, while x_1 , x_3 , and x_4 are corresponding to a pivot (1, -1, and 3, separately). They are the **basic variables**.

It is fine to have free variables, but make sure to order the rows that the appearance of other non-zero pivots are $\searrow$.

Think: when may a consistent system have infinitely many solutions?

3 Existence, Uniqueness, and Gaussian Elimination

3.1 Fundamental Questions in Linear System

We said that the most important thing we care about in linear algebra is to solve the system towards solution sets.

But it is more important to know beforehand whether the system even has a solution:

- 1) Is the system consistent, so that a solution exist?
- 2) If a solution exists, is it unique?

Key Observations.

- All zero row $\Rightarrow$ inconsistent system.

$$\left[\begin{array}{cccc|c} 0 & 0 & 0 & 0 & b \end{array} \right].$$

- Free variable $\Rightarrow$ infinitely many solutions.
- One unique solution $\Leftarrow \text{num}(\text{var}_{\text{lead}}) = \text{num}(\text{var}_{\text{total}})$.

3.2 Gaussian Elimination

In math, we care about a solution, but also care about "the" solution that is unique, minimal, or simplest.

Row echelon form is not unique. That is the issue. For you folks, if I ask you to reduce to row echelon form, my TA and I will get mad — if I really did ask you to do a row echelon form, you can torture me with millions of ways.

$$\left[\begin{array}{cccc|c} 100 & 0 & 200 & 100 & 400 \\ 0 & 0 & -1.414 & 2.828 & 4.242 \\ 0 & 0 & 0 & 81930 & 27310 \end{array} \right]$$

This $\uparrow$ is a row echelon form!!!!

To make our lives easier, we have Gauss-Jordan Elimination to minify this echelon form. And it is super easy, because it is exactly how you write the solution like

$$\begin{cases} x & = 9, \\ y+ & = 1, \\ z & = 6 \end{cases}$$

It creates a fancy matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 9 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 6 \end{array} \right]$$

3.3 Reduced Row Echelon Form

A reduced row echelon form satisfies:

- It is in row echelon form.
- The first nonzero (leading) entry in each nonzero row is 1.
- Each leading 1 is the **only** nonzero entry in its **column**. (For each pivot, above and below are all zeros)

How to get reduced row echelon form

- Step 1: reduce to row echelon form
- Step 2: scale each pivot row to make the leading entry 1
- Step 3: eliminate each column creating zeros above and leftwards

Think: when could reduced echelon form ever have a column containing numbers that is neither 0 nor 1?

3.4 Let's Formalize "TALL" and "WIDE"

Recall that we mentioned "Talllll" and "Wiiide" systems.

Overdetermined : the number of equations is more than the number of unknowns. It is a "Tall" system.

Underdetermined : the number of equations is less than the number of unknowns. It is a "Wide" system.

Think: must an overdetermined system be inconsistent?
Must an underdetermined system have infinitely many solutions?
Is it possible for underdetermined system to have a unique solution?

3.5 Rank of a Matrix

The rank of a matrix A , denoted $r(A)$, is the number of nonzero rows in reduced row echelon form. For $m \times n$ matrix, we have $r(A) \leq \min(m, n)$.

Rank of a matrix is important to determine the existence and uniqueness of solution. For example, given a linear system (I know, this is an inconsistent system):

$$\begin{cases} x + 3z = 1, \\ y + 2z = 0, \\ 0 = 1 \end{cases}$$

We have the Coefficient Matrix A to be

$$\bar{A} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

... and Augmented Matrix $\bar{A}$ to be Given an original augmented matrix $\bar{A}$:

$$\bar{A} = \left[\begin{array}{ccc|c} 1 & 0 & 3 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

Refer to ppt slide no. 81, we know that this is an echelon form because 1) the number of leading 0 of nonzero rows strictly increases (see the downright diagonal shape of zeros to the bottom left?) and 2) all zero rows (if any) are at the bottom, which we do not have.

Refer to ppt slide no. 98, We also know that this is a reduced echelon form because 1) it is an echelon form, 2) the first nonzero entry for each nonzero row is 1, and 3) first nonzero entry in each row is the only nonzero entry in its column.

Also, unfortunately, this is inconsistent, because we observe an ugly row at the bottom with all 0s on the left and a non-zero number on the right. If the non-zero number is 0, then it is consistent. Unfortunately, it is not.

We can also use rank to determine.

First, linear system is **consistent** only when it has no form of $[0 \dots 0 \ b]$, so it is not consistent. In another word, matrix is consistent only if **the rank of augmented matrix $\bar{A}$ equals the rank of coefficient matrix A** .

Second, if a linear system is consistent, the amount of solution depends on **whether there is a free variable**. The number of free variable is determined by the difference between the rank of the coefficient matrix A (rank is 2 because there is an all-0 row) and the augmented matrix $\bar{A}$ (rank is 3 because there is a 1 in the third row of augmented matrix).

If we say $r = r(\bar{A})$ and $n = r(A)$, as $\bar{A}$ is a reduced form, it is always $r \leq n$, and there are $n - r$ free variables.

When summed together:

- If $r = n$, this is the only case to tell A is consistent.
- If $r < n$: this depends. If consistent, then there are infinitely many solutions, with $n - r$ free variables

In this case, $r = 2$ and $n=3$, so even it is consistent, there is one free variable and we have infinitely many solutions.

Beautiful, isn't it?

4 Summary

1. This lecture, we cover the basic ideas of linear systems. We start from the fundamental question of how to solve linear systems (by the way, do you remember what is linear?).

2. To solve a system, we have three pictures to refer to. The row picture is drawing lines of functions. The column picture is to draw vectors. And the matrix.
3. We learn that there are three cases of solutions: one unique, infinitely many, zero. To understand why there can be zero solutions, we introduce the idea of consistency.
4. We learn how to perform elimination on coefficient matrix by reducing it to echelon form. We introduce concepts around pivot.
5. We learned elementary row operation to perform reduction. Nothing too hard, just what we applied with equation systems.
6. We realize that row echelon form has a problem of not being unique, and we explore a way to perform unique result by Gaussian Elimination. The basic idea is to fix each variable's coefficient to 1 and eliminate all possible entries.
7. We discussed possible conditions of row reduced echelon form by applying the Gaussian algorithm, and revisited the consistency of matrices.
8. To further understand the solution sets to systems and why we love systems with N equations and N variables, we learn the rank of a matrix, and learn how to use it to determine the existence and uniqueness of solutions.

As such, this lecture is centered around solutions to linear systems, the existence and uniqueness. We use elementary row operations to manipulate matrix retaining its equivalence. We determine the existence and uniqueness by assessing the consistency. And we learn how to reduce the matrix to the utmost simple form to solve them.

5 In the Next Episode...

We will look at the linear algebra in "column view". We will learn how vectors can be represented, operated, and solved with matrices. We will particularly solve the tricky "multiplication" of vectors, and learn how can an object be transformed in the space under matrix representation.

But before that, good luck with the assignments :3